# DEVON VELLER

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**Lead Game Developer, Artist & Interactive Designer, METIL** **2011 — Present**

*IST — UCF — METIL (Mixed Emerging Technology Integration Lab)*

Multidisciplinary game developer with 14 years bridging art, engineering, and systems design to deliver high-impact interactive training solutions. Executed 300+ products for Y12, Amazon, and Boeing by architecting front-end gameplay systems, back-end infrastructure, and real-time performance optimizations in serious games. Led cross-disciplinary R&D teams leveraging AR/VR/MR technologies, driving innovation in interactive training experiences from concept through deployment while maintaining consistent alignment with client objectives through proactive project leadership.

## EXPERIENCE

**DOE — Senseglove Glovebox Scenario (Year 2)** **2025**

*Delivered | Lead Developer, Technical Artist, & Art Direction*

Led platform migration from HaptX to Senseglove SDK for nuclear glovebox training and simulation, directing art asset implementation and MR integration on VARJO headset for deployment at DOE ORETTC facilities—delivering enhanced haptic training capabilities for critical nuclear operations.

- Engineered Senseglove SDK integration optimizing interaction responsiveness by 30% over previous iterations of Senseglove implementations, resulting in more immersive training validated by DOE personnel.
- Architected performance evaluation system tracking 6+ user interaction metrics with real-time logging, enabling data-driven training assessments while reducing additional PI oversight.
- Co-authored peer-reviewed publication "VR haptics for glovebox operations training" evaluating the efficacy of MR/VR/AR and glove-based haptic training systems for nuclear glovebox operations, contributing technical insights and MR/haptic integration methodologies. frontiersin.org.

Technologies: Unreal Engine, C++, Senseglove SDK, MR (Mixed Reality), VARJO

**DOE — ARTTX - Alarm Response Tactical Training Exercise** **Jan — Sept 2024**

*Delivered | Lead Developer, Art Director, Project Manager*

Led end-to-end development of role-based AR radiological training for 7+ concurrent users on Magic Leap 2, managing project scope, art pipeline, and technical architecture—delivering multi-platform solution praised by DOE client as "This is solid" within 9-month timeline.

- Engineered LLM-assisted development workflow achieving 5.6x faster load times and 25x–6,000x performance improvements in custom pathfinding and decision systems, reducing overall development time by 25%.
- Architected a custom multidimensional spatial partitioning pathfinding solution for small scale environments, operating at 1.3ms on average through all scenario command calls and quantities; enabling real-time navigation for 20+ simultaneously generated character paths.
- Implemented multiplayer using NGO and utilized Unity Authentication, Lobby, and Relay services supporting 7+ concurrent users with `<10ms` latency across low-power AR devices and desktops.
- Engineered a dynamically adjustable render texture camera scheduler for real-time texture rendering distribution, enhanced performance by 320% over the default render texture system.
- Directed art pipeline reusing Digital Twin Interactive Map assets, reducing art production by 40% through strategic asset repurposing and optimization for AR rendering constraints.
- Designed complete UI/UX system with backend service integration, improving user experience by 60% measured through reduced error rates and streamlined communication workflows.
- Built a no-code scenario authoring framework using a custom command system for narrative triggers and events, empowering non-technical designers to create scenarios without engineering support.
- Maintained client alignment through weekly progress reviews and proactive communication, ensuring project deliverables met DOE requirements and security protocols.

Technologies: Unity, OpenXR, C#, NGO (Netcode for GameObjects), Unity Services, AR, Magic Leap 2

### DOE — SMART Interactive Scenario **2023**

*Delivered | Lead Developer, Game Designer & Lead Artist*

Conceptualized and designed an interactive AR radiological training system leading a cross-functional team of 4+ developers and artists to create a customer-facing web portal for real-time Unreal Engine scenario authoring and editing.

- Architected WebSocket backend with JSON serialization enabling zero-latency synchronization between web portal and Unreal Engine, supporting real-time scenario updates for radiation awareness training across distributed teams.
- Designed AI-powered asset generation pipeline reducing 3D cityscape creation time by 25% through automated depth map conversion, scaling content production for customer-facing deployment.
- Created complete UX/UI design system including scenario selection interface and real-time iconography and interactions deployed on iOS/Android AR platforms.
- Led React-based web portal frontend enabling non-technical users to create, edit, and manage interactive scenarios with real-time Unreal Engine synchronization enabling scenario authorship to instruction design team.

Technologies: Unreal Engine, C++, WebSockets, AI Asset Generation Pipelines, iOS/Android

### DOE — HaptX Glovebox Scenario (Year 1) **2022 — 2023**

*Delivered | Technical Artist, Lead Developer, & Art Direction*

Led development and art implementation for high-fidelity nuclear glovebox supporting multiplatform VR collaboration and HaptX haptic integration—delivering physics-accurate training for critical nuclear facility glovebox operations.

- Engineered physics-based torque wrench VR tool with HaptX haptic feedback and using a guided AI workflow, reducing development iteration time by 20% while achieving realistic torque simulation.
- Developed and implemented multiplayer supporting 4 concurrent users and real-time physics with `<100ms` latency across 2 simultanious HaptX glove devices, Vive Focus 3, and Meta Quest—enabling collaborative training in high consequence virtual environments.
- Designed and led the development of competitive glovebox serious game scenario with real-time scoring and displayed performance metrics, increasing collaboration and engagement resulting in securing funding for year 2.

Technologies: Unreal Engine, C++, HaptX SDK, Multiplayer

### DOE — Digital Twin Interactive Map **2022 — 2023**

*Delivered | Lead Game Developer & Technical Artist, Art Direction*

Developed end to end, UX/UI design and art direction for real-time digital twin of DOE ORETTC facility during active construction, designed automated spatial data pipeline and interactive navigation system—enabling stakeholders to visualize and navigate evolving facility architecture.

- Engineered automated data pipeline ingesting architectural documents to generate spatial information and dynamic structure labels, enabling real-time facility updates throughout construction cycle without manual asset recreation.
- Developed dynamic waypoint navigation system generating accessible routes between any 2 rooms across 2 floors with accessibility options, deployed on iOS/Android and windows platforms.
- Created selectable room information data pipline including dynamic room and instructor scheduling, room function data, and carousel image support.
- Directed marketing content creation for DOE ORETTC facility promotion materials, producing 5+ high-impact renderings and animations showcasing interactive features for stakeholder engagement.
- Led asset optimization for real-time rendering on low-power devices and marketing rendering pipelines, achieving consistent 60 FPS performance on iOS/Android platforms through defined texture and polygon budget management.

Technologies: Unreal Engine, C++, Data Pipeline, Pathfinding Algorithms, Blueprint, iOS/Android

**VA — VAPPE** **2021 — 2023**

*Delivered | Lead Front-End Developer & Art Director*

Directed visual design and developed front-end for real-time PPE training application supporting 60 scenarios across iOS and WebGL, establishing VA's new standard for 508-compliant interactive training deployed on Apple Store.

- Architected front-end framework with CSV-driven scenario ingestion supporting real-time remote updates, enabling rapid content deployment during COVID-19 health crisis without app resubmission.
- Directed complete visual identity and branding ensuring consistency across 60 scenarios with 46+ equipment models, establishing design standards compliant with VA accessibility and visual communication requirements.
- Engineered real-time character system with dynamic equipment/clothing skinning driven by scenario data, enabling dynamic procedural character alteration across 46+ posable equipment elements.
- Designed no-code scenario authoring tool using excel and CSV templates, empowering SMEs to create 60 scenarios with 30-60 step training sequences without engineering support—accelerating content creation during health crisis.
- Delivered production-ready application supporting 60 scenarios with integrated narration and step-by-step guidance, achieving 508 compliance deployment on iOS App Store and VA servers.
- Directed art asset pipeline optimizing 46+ blended 3D models and textures for mobile/WebGL performance, optimized for a wide range of character animations while maintaining visual fidelity in both rendered and real-time content each validated by SME provided by VA.

Technologies: Unity, C#, iOS, WebGL, CSV Parsing, 508 Compliance

## EDUCATION

**University of Central Florida, Bachelor of Fine Arts** **2008 — 2011**

**Palm Beach State College, Associate of Arts** **2008**

## TECHNICAL SKILLS

- **Game Engines**: Unreal Engine, Unity, OpenXR, Monogame
- **Programming Languages**: C# (Expert), C++ (Intermediate), JavaScript (Basic), Python (Basic)
- **3D Art & Animation**: Blender, Maya, zBrush, Substance
- **2D Art & UI Design**: Affinity Suite, Photoshop, Illustrator, After Effects
- **Version Control & Development Tools**: Git, Perforce, Jira, Visual Studio, Unity Cloud Services

## ACHIEVEMENTS

- Delivered 300+ products for Y12, Amazon, Boeing and more
- Published applications on Apple Store, most recently VAPPE
- 14 years of professional game development experience